

Identity Gradient (Structural Edition)

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Abstract

Identity Gradient formalises how identity structures evolve under bounded reasoning conditions. It defines the structural grammar governing identity formation, identity deformation, gradient steepening, gradient collapse, and gradient drift. This edition presents the basis-independent formulation suitable for academic publication and cross-domain application.

1 Introduction

Identity Gradient describes how a system’s identity—its stable structural signature—changes as information density, recurrence stability, and boundary motifs evolve. An identity gradient is a directional structural change in identity space.

This edition defines:

- Identity Formation
- Identity Deformation
- Gradient Steepening
- Gradient Collapse
- Gradient Drift
- Gradient Interaction

Identity Gradient is domain-agnostic: it applies equally to reasoning engines, mathematical systems, physical models, and simulation architectures.

2 Identity Formation

Identity forms when structural motifs stabilise into a persistent configuration. Let M denote motif configuration and S structural stability.

Identity forms when:

$$S(M) > S_{\min}.$$

Identity radius G is defined structurally as:

$$G = f(S(M)),$$

where f is a monotonic structural mapping.

2.1 Structural Conditions

Identity forms when:

1. Motif recurrence stabilises.
2. Compression gradients flatten.
3. Boundary information becomes self-consistent.

3 Identity Deformation

Identity deformation occurs when structural motifs lose stability.

Deformation magnitude D is defined as:

$$D = \frac{dG}{dt}.$$

Deformation is triggered by:

- Excessive motif load
- Boundary instability
- Gradient steepening
- Recurrence imbalance

4 Gradient Steepening

Gradient steepening describes increasing directional change in identity space.

Steepening coefficient C is defined as:

$$C = \frac{d^2G}{dt^2}.$$

Steepening increases when:

- Information density rises
- Recurrence capacity decreases
- Boundary motifs become non-linear

5 Gradient Collapse

Gradient collapse occurs when identity can no longer maintain structural continuity.

Collapse condition:

$$C < 0.$$

Collapse velocity:

$$V_c = -\frac{dG}{dt}.$$

Collapse is triggered by:

- Motif overload
- Structural instability
- Recurrence failure

6 Gradient Drift

Gradient drift describes movement of identity in structural space.

Drift velocity:

$$V_d = \frac{dG}{dt}.$$

Drift occurs when:

- Information density changes
- Recurrence capacity evolves
- Boundary motifs reconfigure

7 Gradient Interaction

Multiple identity gradients interact through structural interference.

Interaction strength S :

$$S = g(G_1, G_2, d),$$

where d is separation distance.

Interactions include:

- Gradient merging
- Gradient repulsion
- Gradient resonance

8 Conclusion

Identity Gradient provides the structural grammar governing identity evolution in bounded reasoning systems. It defines how identity forms, deforms, collapses, drifts, and interacts. This edition presents the basis-independent formulation suitable for academic publication and cross-domain application.